

Toward a General Theory of Interaction

Interaction

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Foundational research paper

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Abstract

Interaction is invoked across human-computer interaction, cybernetics, ecology, cognitive science, communication, control, formal methods, robotics, and multi-agent systems, but these traditions do not share a single definition. Some identify interaction with reciprocal action, some with communication, some with feedback, some with synchronized events, and some with situated meaning-making. This paper tests whether interaction is primitive or reducible to established constructs. It concludes that interaction is not a primitive substance or basic event type. It is a higher-order relation over coupled processes: two or more distinguishable systems interact during an episode when changes in each system's possible or actual state trajectory are counterfactually dependent on perturbations transmitted through a coupling relation within a declared boundary and timescale. Reciprocity may be asymmetric, delayed, mediated, probabilistic, or realized across a sequence; simultaneous action, intention, communication, semantic understanding, and visible behavior are not necessary. A single unilateral causal effect is insufficient unless it enters a closed or recurrent dependency structure. Existing fields provide nearly all constituent concepts - action, coupling, feedback, transition, synchronization, communication, coordination, and situated interpretation - but no single theory supplies a cross-domain identity, composition, and comparison framework. The warranted contribution is therefore a minimal synthesis, not a new primitive: a typed interaction ontology, episode lifecycle, compositional grammar, algebraic operators, identity criteria, invariants, interaction contract, and falsification matrix.

1. Problem and Method

The word interaction often performs explanatory work without identifying what kind of phenomenon occurred. In interface research it may mean a sequence of user inputs and system outputs. In cybernetics it can imply feedback. In communication theory it may mean exchange of messages. In ecological psychology it can indicate organism-environment reciprocity. In formal computer science it can mean synchronization or communication between processes. In social theory it may refer to jointly produced order. These uses overlap, but none can be assumed to define the others.

The analysis used a cross-disciplinary boundary method. For each tradition, the study identified: (1) the entities allowed to interact; (2) the relation or mechanism linking them; (3) whether reciprocity, intention, communication, observation, or state change is required; (4) the temporal unit; (5) the evidence by which interaction is established; and (6) the formal resources available for identity and composition. Candidate definitions were then tested against counterexamples including passive sensing, unilateral control, delayed feedback, hidden API effects, failed communication, and simultaneous but independent activity.

The supplied corpus was used as foundational evidence for ecological, situated, distributed, activity-theoretic, cognitive-engineering, and interaction-design perspectives. The methodological standard followed the Research Canon: concepts were retained only when necessary, definitions were separated from implementations, and major claims were made falsifiable and operationalizable.

2. Existing Conceptions of Interaction

No accepted universal definition exists. The major traditions organize interaction around different explanatory targets.

Ecological psychology treats organism and environment as complementary rather than independent explanatory domains. Gibson's affordances are possibilities of action relative to an animal-environment system. This supplies relationality but does not by itself define an interaction episode, identity, or composition.

Cybernetics organizes control and communication through feedback. Wiener placed messages and responses within circular regulation. Feedback gives a strong mechanism for recurrent interaction, but open-loop influence and one-shot synchronized events show that feedback is not necessary for every interaction token.

Systems theory commonly treats interaction as mutual influence among components whose collective behavior is not recoverable from isolated components alone. This supplies dependence and emergence but often leaves interaction boundaries and empirical identity underspecified.

Situated action and ethnomethodological approaches reject the view that plans fully determine conduct. Suchman describes successful interaction as collaboratively produced intelligibility responsive to contingencies and others' actions. This supplies situated accomplishment and interpretive coordination, but it is too restrictive for non-semantic physical and computational interactions.

Distributed cognition locates cognitive organization across people, artifacts, and representational media. Hutchins shows that coordinated transformations across a system can constitute the relevant cognitive process. This supports system-level interaction and mediation but makes boundaries a substantive modeling decision.

Activity theory places action within object-oriented, tool-mediated, socially organized activity. It captures historical and institutional mediation, yet its unit is often too large for atomic computational interaction.

HCI contains multiple theories rather than one definition. Cognitive engineering models user-system exchanges quantitatively; embodied interaction treats meaning as arising through engaged practice; instrumental and generative theories organize design around action and adaptation; computational rationality explains interaction as adaptation under bounds. These theories explain important classes of interaction but are not extensionally equivalent.

Communication theory distinguishes transmission from meaning. Shannon's mathematical theory explains signal selection and uncertainty without requiring semantic understanding. Speech-act and agent-communication theories add illocutionary force and protocol commitments. Communication is therefore one interaction mechanism, not interaction's genus.

Control theory defines systems through inputs, states, outputs, and dynamical response. Coupling occurs when one subsystem's output enters another's dynamics. This gives precise counterfactual and temporal machinery, but interaction cannot be reduced to successful control because uncontrolled perturbation and negotiation also qualify.

Formal methods represent interaction through synchronization, shared events, message passing, handshakes, shared variables, or transitions. CCS, CSP, process algebras, labeled transition systems, event structures, and protocol models provide composition and behavioral equivalence. Their strength is formal precision; their limitation is that semantic and social interpretation normally lies outside the transition semantics.

Robotics and multi-agent systems treat interaction through sensing, action, communication, coordination, negotiation, and joint task execution. Protocols specify admissible message sequences and roles. These approaches show that interaction may be globally specified but locally enacted, and that composition can fail when local agents cannot realize a global protocol.

Philosophy of action distinguishes intentional action from mere movement or event. This blocks a definition of interaction as simply two actions: many interactions include non-agentive components, unintended consequences, or automatic responses.

3. Reductive Tests

Interaction cannot be reduced to action. An action can occur without another system being coupled to it, while an interaction can include passive dynamics, automatic responses, or non-intentional perturbations. Action is attributable to an agent under some theory of agency; interaction concerns dependence among processes.

Interaction cannot be reduced to behavior. Behavior is an observed or modeled pattern attributable to a system. Two systems can behave simultaneously without interacting, and a system's behavior may be evidence of interaction without constituting the relation itself.

Interaction cannot be reduced to communication. Mechanical contact, resource contention, sensorimotor coupling, and shared-state interference can be interactions without messages or shared meaning. Conversely,

a transmitted message that produces no uptake may constitute attempted communication but may fail to establish an interaction episode under a causal-dependence criterion.

Interaction cannot be reduced to computation. Computation can occur internally without coupling to another system. Some interactions are physical, biological, or social and are not usefully described as computation. Computational interaction is one domain-specific realization.

Interaction cannot be reduced to observation. Observation may be unilateral and non-perturbative at the modeled scale. It becomes interaction when sensing changes subsequent trajectories, when measurement perturbs the observed system, or when observation participates in a recurrent loop.

Interaction cannot be reduced to a state transition. A state transition can be internal and autonomous. Interaction requires that at least part of a transition's occurrence, probability, timing, or reachable consequences depend on another distinguishable system through a coupling.

Interaction cannot be reduced to causation in general. A falling rock may cause a mark in soil, but a single one-way causal event does not automatically justify an interaction episode. The term is most useful when the coupling creates reciprocal or recurrent constraint across the participants' trajectories. Reciprocity need not be equal or immediate; it may be mediated through a shared environment and realized over time.

These failures do not establish interaction as metaphysically primitive. They establish it as a non-redundant analytical relation composed from existing notions of system, state, process, coupling, perturbation, constraint, and temporal dependence.

4. Canonical Definition

Interaction episode: a bounded interval in which two or more distinguishable systems are coupled such that perturbations originating in, or mediated by, one system alter the possible or actual state trajectory of at least one other system, and the resulting configuration constrains a later trajectory of an originating or co-participating system.

This definition has six commitments.

First, interaction is relational. It is not located exclusively in either participant.

Second, interaction is processual. It is extended over at least an analytically distinguishable sequence, even when implemented as an apparently atomic synchronization.

Third, interaction requires coupling. A pathway must exist through which perturbation, constraint, resource competition, synchronization, signal, or material effect can pass.

Fourth, interaction requires counterfactual dependence. Holding relevant background conditions fixed, removing or changing the coupling or participant would change some participant's reachable, probable, timed, or actual trajectory.

Fifth, interaction requires reciprocal closure at the episode level. The reciprocal effect can be direct, indirect, asymmetric, delayed, probabilistic, or environmentally mediated. A single isolated one-way influence is causation but not yet interaction under the strict definition.

Sixth, every interaction claim is boundary- and timescale-relative. What appears unilateral over milliseconds may be reciprocal over seconds; what appears dyadic may be mediated by infrastructure, institutions, or environments when the boundary expands.

5. Necessary and Non-Necessary Conditions

Necessary conditions are: distinguishable participants or subsystems; a coupling relation; at least one transmitted perturbation or constraint; counterfactual effect on a participant trajectory; reciprocal or recurrent closure within the declared episode; and explicit boundary and timescale.

Not necessary are: biological agency; conscious intention; semantic content; successful communication; mutual awareness; equality of influence; simultaneous response; visible interface events; goal agreement; positive outcome; or permanent state change.

Observable state change is not strictly necessary if the interaction changes transition probabilities, timing, control margins, uncertainty, or reachable possibilities without producing an immediately observed macrostate change. Such claims require stronger evidence than direct transition traces.

Interaction may fail. A protocol can deadlock, a conversational repair can collapse, or a control loop can oscillate. Failure does not erase the interaction; it changes its outcome and lifecycle state.

6. Interaction Ontology

The minimal ontology contains established constructs rather than newly coined entities.

Participant: a system or subsystem treated as distinguishable for the analysis.

State: the variables considered sufficient to characterize a participant at a time and scale.

Process: an ordered or partially ordered evolution of state.

Boundary: the inclusion rule separating participants, environment, and background conditions.

Coupling: a relation through which one participant or shared medium can affect another participant's transition structure.

Perturbation: a difference propagated through a coupling relation.

Constraint: a change in possible, probable, timed, or admissible trajectories.

Uptake: incorporation of a perturbation into a participant transition or control process. Uptake need not be semantic.

Response: a subsequent transition attributable partly to uptake.

Recurrence: return influence through which a response constrains an earlier source, co-participant, or shared coupling configuration.

Episode: a bounded segment containing initiation, coupling, uptake, response, and sufficient closure for identity.

Trace: a temporally ordered or partially ordered record of observable events and state estimates.

Protocol: a rule system constraining admissible interaction traces.

Outcome: the episode-relative resulting state, distribution, commitment, or termination condition.

Context: background conditions that affect coupling or interpretation but are not modeled as participants in the focal episode.

Environment: modeled external dynamics through which participants may be coupled.

Invariant: a declared property preserved across an interaction, representation, implementation, or decomposition.

7. Interaction Primitives and Grammar

The theory does not claim metaphysical primitives. It proposes analytical primitives sufficient for representation.

A minimal event vocabulary is:

bind(p, q, c): establish or activate coupling c between participants p and q.

emit(p, d): participant p produces perturbation d.

transmit(c, d): coupling c carries d.

uptake(q, d): d changes q's transition conditions or state.
 respond(q, r): q undergoes or produces response r.
 recur(r, p|c): r constrains p or the coupling configuration.
 commit(o): an outcome or commitment becomes episode-relevant.
 fail(f): a declared failure condition occurs.
 unbind(c): coupling terminates or becomes inactive.

A minimal interaction grammar is:

Episode ::= Boundary Initiation CoupledProcess Closure
 CoupledProcess ::= Exchange | Synchronization | SharedMedium | Feedback | Coordination
 Exchange ::= Emit Transmit Uptake [Response]
 Synchronization ::= JointEvent constrained-by Participants
 SharedMedium ::= ParticipantEffect MediumTransition ParticipantUptake
 Feedback ::= Exchange Recurrence+
 Coordination ::= CoupledProcess constrained-by Protocol
 Closure ::= Recurrence | JointCommitment | StableTermination | DeclaredTimeout | Failure

The grammar separates implementation form from interaction identity. A message exchange, button press, physical collision, or shared-memory write may instantiate the same abstract roles while differing in semantics and mechanism.

8. Interaction Lifecycle

1. Eligibility: participants, capacities, permissions, and preconditions make coupling possible.
2. Initiation: a participant or environmental condition activates a coupling pathway.
3. Perturbation: a difference is produced and propagated.
4. Uptake: another participant's state or transition structure becomes dependent on that difference.
5. Response: a consequential transition occurs.
6. Recurrence or closure: the response returns influence, establishes a joint commitment, reaches stable termination, times out, or fails.
7. Aftereffect: participant states, beliefs, policies, coupling strength, or future opportunities may change.

These phases are analytical, not universally sequential. Synchronization can merge initiation, uptake, and response. Distributed interaction can contain concurrent branches. Repeated interaction can overlap aftereffects with the eligibility conditions of later episodes.

9. Identity and Invariants

Interaction identity is not exhausted by event equality. The same interaction may be rendered through different modalities, implementations, or trace granularities. Conversely, superficially identical event sequences may constitute different interactions when participants, semantics, authority, or causal dependencies differ.

An interaction identity claim must declare: participants and roles; focal boundary; timescale; coupling topology; relevant state variables; protocol or admissibility conditions; causal or counterfactual dependencies; initiation and closure criteria; and invariants.

Strong token identity requires continuity of the same causal episode. Type identity requires equivalence under a declared abstraction.

Candidate invariants include:

Participant-role invariance: the same abstract roles are preserved despite implementation changes.
 Protocol invariance: admissible ordering and obligation structure are preserved.
 Transition invariance: corresponding actions induce equivalent abstract state changes.
 Causal-structure invariance: the counterfactual dependency graph is preserved.

Outcome invariance: declared terminal conditions or distributions are preserved.

Information invariance: selected distinctions available to participants are preserved.

Opportunity invariance: relevant feasible, permitted, exposed, and recoverable actions are preserved.

Behavioral invariance: trace distributions remain equivalent under a metric and tolerance.

No invariant is universal. Every equivalence claim must name the preserved relation, observer, abstraction, context, and tolerance.

10. Interaction Algebra

Let an interaction type I be represented as a tuple:

$$I = \langle P, S, C, E, T, R, B, K \rangle$$

where P is a set of participants and roles; S is a structured state space; C is a coupling topology; E is a typed event set; T is a transition or stochastic kernel; R is a recurrence and dependency relation; B specifies boundary and timescale; and K specifies closure conditions and invariants.

The following operators are sufficient for a minimal compositional algebra.

Sequential composition $I ; J$: J begins after a closure condition of I and may inherit a mapped state.

Parallel composition $I \parallel J$: episodes proceed concurrently with independent or shared participants.

Synchronization $I \text{ sync_} A J$: episodes must jointly participate in events from synchronization set A .

Choice $I + J$: one of several admissible episode structures is realized.

Iteration I^* : zero or more recurrent instances under a termination rule.

Mediation $M[I]$: interaction I is realized through mediator or environment M .

Restriction $I \backslash H$: selected events or coupling channels are hidden from an observer while remaining causally active.

Abstraction $\alpha(I)$: implementation events are mapped to a coarser interaction type.

Refinement $I \leq J$: J preserves the declared invariants and allowed behavior of I while adding implementation detail.

Interference $I \bowtie J$: concurrent interactions share state or resources such that each changes the other's reachable traces.

These operators borrow from process algebra, transition systems, and systems theory. The proposed synthesis adds explicit participant roles, episode boundaries, recurrence criteria, and cross-domain invariants. Algebraic laws are conditional rather than universal. Sequential composition is not necessarily commutative; parallel composition may cease to be associative when resource or timing assumptions change; abstraction must preserve declared observables to support equivalence.

11. Composition Rules

Interactions compose only when boundary mappings are valid. A participant in one episode may become an environment, mediator, or internal subsystem in another abstraction.

State compatibility: the terminal state distribution of the first episode must satisfy the initial-state assumptions of the second.

Role compatibility: inherited participant roles and authority must remain meaningful.

Coupling compatibility: channels, shared variables, or physical interfaces must connect without introducing unmodeled transformations.

Temporal compatibility: ordering, concurrency, deadlines, and latency assumptions must be satisfiable.

Semantic compatibility: where messages or commitments are involved, content interpretations and protocol meanings must align.

Invariant compatibility: composition must not silently violate safety, identity, or observational equivalence conditions.

Closure discipline: a subinteraction may be incomplete while the containing interaction remains active, but its completion, timeout, cancellation, or compensation must be represented.

Composition is therefore not mere concatenation of event logs. It is a constrained construction over state, roles, coupling, time, and invariants.

12. Comparative Cases

Human-human conversation. Participants alternate and overlap utterances, gestures, gaze, repair, and inference. Communication and semantic interpretation are central, but the interaction also includes timing, turn allocation, mutual orientation, and embodied feedback. Identity depends on the evolving joint episode rather than isolated messages.

Human-computer interaction. A user action changes system state or presented information; the resulting projection changes the user's opportunities, beliefs, or next action. Reciprocity is asymmetric because the computational system need not possess intention. Interaction is established by coupled trajectories, not by anthropomorphic agency.

Agent-agent interaction. Agents exchange messages, synchronize on shared state, negotiate, or coordinate through protocols. Interaction can be formally represented as global protocols and local enactments. Failed enactability demonstrates that a global trace specification is not sufficient unless local participants can realize it.

API interaction. A client request invokes a service transition, and the response, error, side effect, or timeout changes the client's later state or policy. A fire-and-forget request with no later client dependence may be a unilateral invocation; an episode becomes interaction when returned or shared-state effects constrain subsequent behavior.

Physical interaction. Two bodies collide, deform, transfer momentum, and alter each other's trajectories. There is no required communication or interpretation. Reciprocal physical dependence provides the clearest minimal example.

Conversational interaction with a computational agent. Messages are represented as linguistic acts, but the underlying episode also includes model state, tool availability, authorization, latency, and projection. Apparent semantic reciprocity can exceed actual computational capability, so interaction analysis must separate signaled, believed, and feasible transitions.

Across all six cases, what remains common is not intention, language, interface, or computation. It is bounded reciprocal constraint through coupling, observable or inferable in participant trajectories.

13. Evidence and Operationalization

Direct evidence includes synchronized event traces, state measurements, messages, control signals, resource changes, and observed transitions. Derived evidence includes reconstructed causal graphs, reachability differences, transfer functions, protocol conformance, and temporal dependencies. Interpretive evidence includes inferred intention, meaning, belief, coordination, and social significance.

A minimal empirical test for interaction compares a coupled condition with a suitably matched decoupled or perturbed condition. Evidence should show that changing participant A or coupling C changes B's trajectory and that B's resulting state or response changes a later trajectory of A, C, or a shared system within the episode boundary.

Possible measures include conditional mutual information, intervention effects, transfer entropy, Granger-style temporal dependence, impulse response, synchronization, cross-recurrence, state-transition divergence, protocol trace conformance, and causal model comparison. None alone proves interaction. Statistical dependence without a plausible coupling may be confounding; causal influence without recurrence may be unilateral causation; synchronized traces may share an external driver.

For semantic and social interaction, additional evidence may include repair sequences, uptake displays, commitment changes, alignment, or coordinated task transitions. Such evidence should not be generalized to physical or non-semantic interaction.

14. Falsification Tests

The theory should be rejected or narrowed if any of the following is demonstrated.

Reduction to transition systems: if standard transition-system semantics can represent participants, coupling, recurrence, episode identity, interpretation boundaries, and cross-domain composition without additional commitments, the proposed ontology adds no explanatory value.

Reciprocity failure: if scientific practice consistently classifies one-way causal influence as interaction and no explanatory or predictive gain follows from distinguishing it, reciprocal closure should be removed from the definition.

Boundary non-reproducibility: if independent researchers cannot agree on episode boundaries and timescales under explicit rules, interaction identity may be too observer-relative for stable comparison.

Counterfactual non-identifiability: if coupled and merely correlated trajectories cannot be distinguished even in principle for relevant domains, interaction cannot support causal explanation there.

Composition failure: if the algebra cannot predict or explain properties of composed episodes beyond raw event concatenation, it should be treated as notation rather than theory.

Invariant failure: if no useful property can be preserved across implementation-independent representations of interaction, claims of general interaction types should be abandoned.

Existing-theory sufficiency: if one established theory provides equivalent vocabulary, ontology, mechanisms, formalization, and empirical predictions across all comparative cases, the synthesis should be replaced by that theory.

Predictive null: if interaction-level variables do not improve explanation or prediction over participant-level states, independent causal models, and background context, interaction may be merely descriptive.

15. Threats to Validity

The definition privileges reciprocal constraint and may exclude ordinary language uses such as a person interacting once with a passive artifact. This is deliberate but contestable. The stricter definition seeks explanatory discrimination; broader usage may be appropriate in design practice.

Counterfactual dependence is model-relative. Incorrect state variables, hidden common causes, or poorly chosen timescales can create or erase apparent interaction. Reciprocal effects may be too weak or delayed to estimate.

The framework combines traditions with incompatible ontological assumptions. Ecological psychology resists internal representational explanations; formal computer science abstracts away material embodiment; social theory treats meaning and accountability as locally accomplished. The synthesis claims a shared structural layer, not theoretical unification of all explanatory levels.

Formal composition can obscure situated transformation. Real participants reinterpret protocols, change goals, and alter boundaries. Interaction types should therefore be treated as models with scope conditions, not exhaustive descriptions.

The comparative cases are selected to maximize diversity, but they do not establish universality. Biological symbiosis, market interaction, quantum interaction, and institutional interaction may require revisions.

16. Conclusion

Interaction is not a primitive substance, action type, or universal mechanism. Nor is it adequately reducible to action, behavior, communication, computation, observation, state transition, feedback, or causation alone.

The literature supports a minimal cross-disciplinary synthesis: interaction is a bounded, temporally organized relation of reciprocal constraint among distinguishable systems through coupling. Its tokens are established by counterfactual dependence in participant trajectories; its types are defined only under declared boundaries, roles, abstractions, and invariants. Communication, synchronization, feedback, coordination, collision, shared-state interference, and situated meaning-making are mechanisms or forms of interaction rather than synonyms.

Existing disciplines provide the constituent theory, but no single one provides a complete cross-domain account of identity, composition, invariance, and evidence. The minimum conceptual addition is therefore not a new primitive. It is a shared interaction contract and typed compositional framework that makes existing concepts interoperable and empirically testable.

17. Reusable Artifacts

Interaction Taxonomy

Element	Definition / specification
By coupling mechanism	physical contact; field or environmental mediation; signal transmission; shared state; resource contention; synchronization; institutional or normative mediation
By temporal form	atomic synchronization; request-response; feedback loop; recurrent exchange; continuous coupling; asynchronous protocol; episodic coordination
By agency	non-agentive; unilateral-agentive; dyadic agentive; multi-agent; human-machine; system-environment
By semantic status	non-semantic; informational; communicative; interpretive; commitment-forming
By symmetry	symmetric; asymmetric; leader-follower; controller-plant; client-service; many-to-many
By outcome	successful; failed; partial; deadlocked; oscillatory; compensated; abandoned

Interaction Contract

Element	Definition / specification
Identity	interaction type and token identifier; abstraction level
Participants	entities, roles, capabilities, authority, and inclusion criteria
Boundary	system/environment partition; timescale; start and closure rules
State	relevant state variables and uncertainty
Coupling	channels, shared media, topology, latency, and

Element	Definition / specification
	capacity
Preconditions	eligibility, permissions, enablement, and initial assumptions
Events	emission, transmission, uptake, response, recurrence, commitment, failure
Transitions	deterministic, nondeterministic, or probabilistic state effects
Protocol	admissible ordering, synchronization, obligations, and timeouts
Invariants	properties required to remain preserved
Evidence	direct observations, derived dependencies, inferred interpretations
Failure and recovery	deadlock, rejection, timeout, compensation, cancellation, rollback
Completion	stable termination, commitment, goal condition, timeout, or declared truncation

Falsification Matrix

Claim	Contrary evidence and consequence
Interaction is relational	Cases where interaction is fully attributable to one participant without coupling -> Remove relationality or restrict domain
Reciprocal closure is necessary	Reliable explanatory practice treating isolated one-way influence as interaction -> Broaden definition to directed causal coupling
Counterfactual dependence distinguishes interaction	Matched interventions fail to separate interaction from common-cause correlation -> Limit causal claims to descriptive coordination
Interaction types survive implementation change	No useful invariants across implementations -> Abandon general types; retain local episode descriptions
Composition adds explanatory value	Composed properties equal unconstrained event concatenation -> Drop algebra; use trace notation only
Synthesis is non-redundant	One established theory matches all ontology, formal, and empirical requirements -> Adopt that theory and reject the synthesis

Dependency Graph

Ontology -> state and participant identity -> coupling -> opportunity -> perturbation and uptake -> interaction episode -> interpretation and policy -> behavior -> evidence and measurement -> comparison and prediction.

Interaction Comparison Protocol

1. Declare the focal boundary, timescale, participants, roles, and state abstraction.
2. Identify coupling mechanisms and construct a counterfactual decoupled baseline.

3. Enumerate observable events, hidden state assumptions, and uncertainty.
4. Test directed influence and reciprocal or recurrent closure.
5. Identify protocol, semantic, causal, outcome, opportunity, and behavioral invariants separately.
6. Compare traces only after aligning state abstraction and event granularity.
7. Control for common causes, shared external drivers, latency, and measurement artifacts.
8. State whether the claim concerns token identity, type equivalence, refinement, or similarity.
9. Report failures, deadlocks, timeouts, and recovery as part of the interaction rather than deleting them.
10. Declare the conditions under which the comparison would be falsified.

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